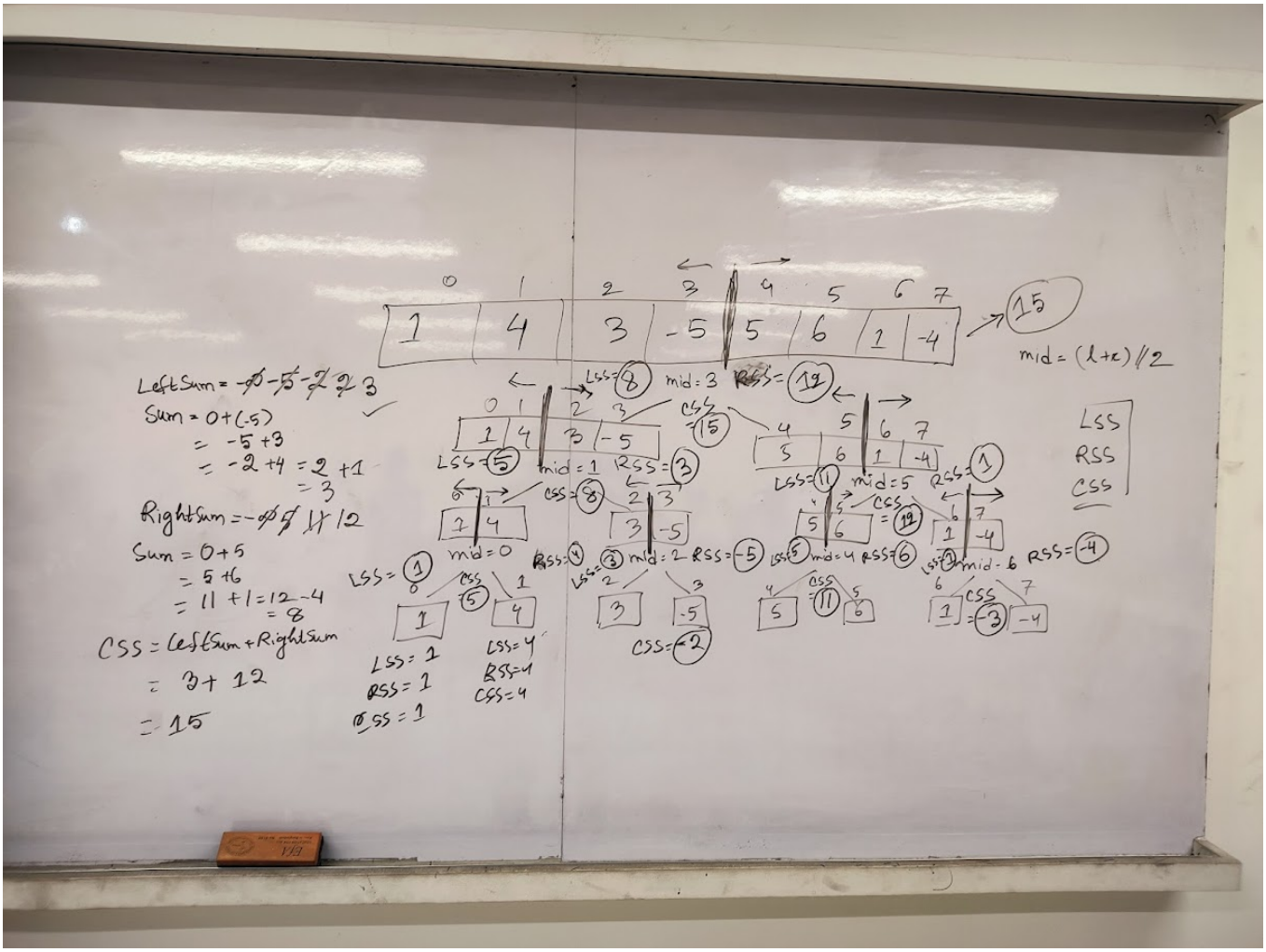
Maximum Sum SubArray



Pseudo Code:

```
public class MaxSubArraySum {
    public static int maxSumSubArray(int[] arr, int start, int end){
        if (start == end)
        {
            return arr[start];
        }

        int mid = (start+end) / 2;

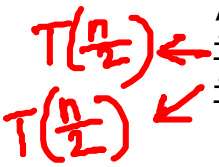
        // Recursively finding max sum in the left and right halves
        int leftSubArraySum = maxSumSubArray(arr, start, mid);
        int rightSubArraySum = maxSumSubArray(arr, mid+1, end);

        // FInding max crossing sum
        int leftSum = Integer.MIN_VALUE;
        int sum = 0;
        for (int i = mid; i >= start; i--)
        {
            sum += arr[i];
            if (sum > leftSum)
            {
                leftSum = sum;
            }
        }

        int rightSum = Integer.MIN_VALUE;
        sum = 0;
        for (int i = mid+1; i <= end; i++)
        {
            sum += arr[i];
            if (sum > rightSum)
            {
                rightSum = sum;
            }
        }

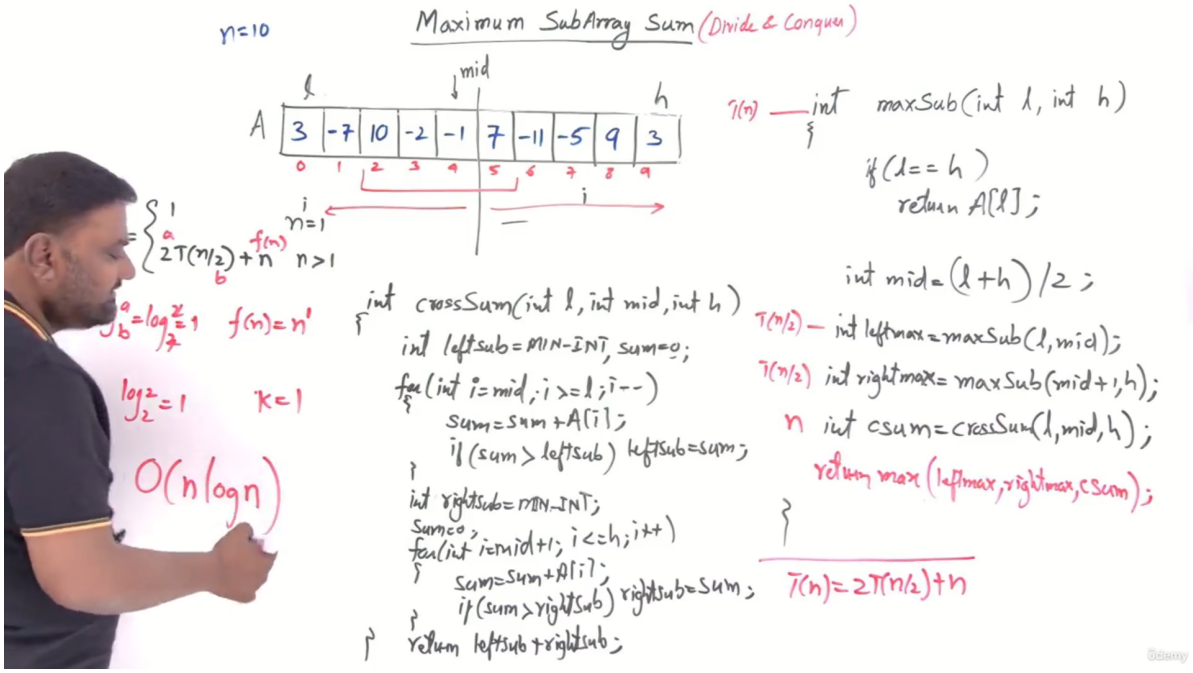
        int crossSum = leftSum + rightSum;

        return Math.max(Math.max(leftSubArraySum, rightSubArraySum),
crossSum);
    }
}
```









$$T(n) = 2T\left(\frac{n}{2}\right) + O(n)$$

$$= 2T\left(\frac{n}{2}\right) + n$$

$$= 2\left[2T\left(\frac{\frac{n}{2}}{2}\right) + \frac{n}{2}\right] + n$$

$$= 4T\left(\frac{n}{4}\right) + 2n$$

$$= 4\left[2T\left(\frac{\frac{n}{4}}{2}\right) + \frac{n}{4}\right] + 2n$$

$$= 8T\left(\frac{n}{8}\right) + 3n$$

$$= 2^3 \cdot T\left(\frac{n}{2^3}\right) + 3n$$

$$\vdots$$

$$k^{th} \rightarrow 2^k \cdot T\left(\frac{n}{2^k}\right) + Kn \quad \text{.....(1)}$$

Substitution Method

$$\frac{n}{2^K} = 1$$

$$\Rightarrow 2^K = n$$

$$\Rightarrow \log_2 2^K = \log_2 n$$

$$\Rightarrow K \cdot \log_2^2 = \log_2 n$$

$$\Rightarrow \boxed{K = \log_2 n}$$

Putting the value in (1) →

$$T(n) = 2^K \cdot T\left(\frac{n}{2^K}\right) + Kn$$

$$= 2^{\log_2 n} \cdot T\left(\frac{n}{2^{\log_2 n}}\right) + n\log_2 n$$

$$= n^{\log_2 2} \cdot T\left(\frac{n}{n^{\log_2 2}}\right) + n\log_2 n$$

$$= nT\left(\frac{n}{n}\right) + n\log_2 n$$

Return করে ← $= nT(1) + n\log_2 n$

$$= n + n\log_2 n$$

$$= O(n\log_2 n)$$